Denis Bolonin
Data Scientist, **5+ years** of experience
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UK full-time work permit, valid until Jan 2028 London, UK

PROFESSIONAL SUMMARY

Software Engineer with **over 5 years** of production experience and hands-on applied ML research, including an MSc in Computer Science (QMUL) and multimodal LLM research with HSE University. Specialized in rigorous model evaluation and benchmarking, parameter-efficient fine-tuning, and multimodal vision-language models, with a strong foundation in statistical analysis. Built and deployed LLM orchestration and OCR-based document automation in production at digital.sfpe.
Actively engaged in the open-source community, I've contributed to projects like ClickHouse, a prominent open-source DBMS, with technical work recognized by the project's CTO.

RESEARCH PROJECTS

Multimodal Semantic Analysis of Scientific Publications via Adaptive Fine-Tuning 2025 - 2026
Research Project, HSE University
Code: github.com/BoloniniD/Multimodal-Scientific-Publication-Analysis

- Built a vision-language system adapting the Russian-specialized YandexGPT-5-Lite-8B model with a CLIP visual encoder to jointly reason over figures and text in Russian-language scientific publications, addressing a gap in predominantly English-centric multimodal ML research.
- Designed FIM-guided surgical fine-tuning: used Fisher Information Matrix scores to selectively apply QLoRA adapters to the most task-relevant 40% of model layers.
- Outperformed standard uniform QLoRA fine-tuning on every metric across two tasks (figure captioning and three-class claim verification); the fine-tuned 8B model outperformed zero-shot Gemma and Qwen baselines on Russian scientific tasks. Tech stack: Python, PyTorch, Hugging Face Transformers, QLoRA, CLIP.

Automated Layer Selection for Efficient Fine-Tuning of Medical Image Segmentation Models 2024 - 2025
MSc thesis, Computer Science, Queen Mary University of London
Code: github.com/BoloniniD/Automated-Layer-Selection-for-Efficient-Fine-Tuning

- Developed a parameter-efficient fine-tuning framework to address catastrophic forgetting in medical image segmentation models across sequential clinical tasks.
- Implemented and benchmarked Surgical Fine-Tuning, Trainable Projected Gradient Method (TPGM), and a novel hybrid approach on the CSWin-UNet architecture across a three-stage sequential learning protocol using abdominal CT datasets.
- TPGM achieved the best trade-off between knowledge preservation and task adaptation; all proposed methods avoided the catastrophic forgetting seen with standard fine-tuning.
- Reduced training time substantially, enabling full experimentation on a single consumer-grade GPU. Built with Python and PyTorch.

TECHNICAL SKILLS

Data Science & Machine Learning: Machine Learning, Deep Learning, Statistical Analysis, Experiment Design, A/B Testing, pandas, NumPy, scikit-learn, Feature Engineering, Data Visualization (Matplotlib), Natural Language Processing (NLP), Computer Vision, PyTorch, Hugging Face Transformers, QLoRA/LoRA, Parameter-Efficient Fine-Tuning (PEFT), Vision-Language Models, CLIP, Model Evaluation, Hyperparameter Tuning, Model Deployment, LLM Integration & Orchestration, Agentic Workflows, RAG Pipelines, Prompt Engineering, OCR, Reinforcement Learning, n8n
Programming Languages: Python, C++, SQL, Rust, JavaScript, C, Bash, R
Backend Development: Django, FastAPI, Flask, REST APIs, gRPC, Microservices
Databases & Data Storage: PostgreSQL, ClickHouse, Redis, SQLite, Data Pipeline Development
Cloud & DevOps: Docker, Kubernetes, OpenShift, Jenkins, Helm, AWS, Istio, CI/CD, Infrastructure as Code
Big Data & Streaming: Apache Kafka, Apache Spark, Hadoop, RabbitMQ, ZooKeeper
Monitoring & Analytics: Prometheus, Grafana, Kibana, Performance Optimization
Testing: pytest, Unit Testing, Functional Testing, Fuzzing
Project Management: Agile, Scrum, Jira, Confluence, Team Leadership, Mentoring

WORK EXPERIENCE

Senior Software Engineer (Data Science & Backend) 08/2024 - Present
digital.sfpe, a software development SaaS startup

- Developed from the ground up a comprehensive software solution tailored for fire protection engineering companies to ensure regulatory compliance. This solution automates critical processes such as contract management, data validation, format control, minimizes errors and proactively monitors deadlines. The implementation of this software significantly improved efficiency, enabling a **single person** to manage the workload that previously required **3-4 full-time employees**.
- Designed and deployed **LLM-powered automation in production**: an LLM-orchestrated product chatbot and OCR-based payment document parsing for the billing pipeline, integrating n8n workflows with custom Python services.
- Implemented autofill process for EMERCOM contract notifications, reducing the processing time per notification from 8-10 minutes to under 1.5-2 minutes (**over 80% speed up**).
- Engineered an end-to-end subscription billing system (order intake, payment document parsing, invoicing, and reconciliation), replacing an almost fully manual workflow and eliminating human error across the billing pipeline.

Software Developer 07/2022 - 07/2024, 2 years 1 month

Sber, one of the biggest fintech companies

- Developed a utility for financial models for a high-value investment project, specifically valued at 2 billion USD, **accelerating** model calculations **by over 50%** through concurrent processing.
- Engineered an automated data pipeline for depositary settlements, performing ingestion, validation, and reconciliation of settlement data and fully eliminating manual intervention and reporting errors.
- Designed team delivery metrics and built an analytics dashboard for Scrum Masters, translating raw sprint activity data into actionable insights and **increasing team throughput by 15%**.

Open Source Developer 10/2020 - 06/2022, 1 year 9 months

ClickHouse, opensource DBMS

- Researched and implemented UDP-based replication protocols (UDT, µTP/uTP, ENet) for ClickHouse, evaluating their performance and laying groundwork for future QUIC protocol studies.
- Integrated Rust into ClickHouse build system, enabling C++ and Rust interoperability. Implemented and integrated the BLAKE3 hashing function, achieving almost **100% performance increase** with a new hashing function compared to existing functions.
- Implemented YAML configuration support for ClickHouse, providing users with a more readable and convenient configuration format.

EDUCATION

Master's degree in Computer Science. Queen Mary University of London, London, UK 09/2024 - 09/2025

Software and Data Engineering stream

Bachelor's degree in Computer Science. HSE University, Moscow, Russia 09/2019 - 06/2023

Applied Mathematics and Informatics, Distributed systems

CERTIFICATIONS

ICAgile Certified Professional 12/2022

Completed Agile training, received the ICAgile Certified Professional

Certificate Link

Applied statistical analysis 05/2022

Completed additional Applied Statistical Analysis course, HSE University

PUBLICATIONS

More Than 2x Faster Hashing in ClickHouse Using Rust

Co-authored technical blog post on integrating Rust into ClickHouse. The work was recognized by ClickHouse CTO Alexey Milovidov, who wrote: “I'm highly impressed by the work of Denis Bolonin that made it possible!” ClickHouse Official Blog

ACHIEVEMENTS

Winner of the Finance block hackathon at Sber 09/2023

- Designed and developed the data layer for a mobile approval platform, including storage and management of user data and incoming approval requests.
- Collaborated with the team on product ideation and workflow design.
- The solution replaced a cumbersome legacy process with a more engaging, gamified approval experience.